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Research paper

Determination of optimal transmission network constraints for novel close-to-real-time flexible energy market

Danka Veljković [a,b,*], Zoran Vujasinović [c], Milan Ivanović [a], Željko Đurišić [b]

[a] Electricity Coordinating Center (EKC), Belgrade, Serbia
[b] University of Belgrade School of Electrical Engineering, Belgrade, Serbia
[c] European Union Agency for the Cooperation of Energy Regulators, Ljubljana, Slovenia

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ABSTRACT

The rapid integration of renewable energy sources, characterized by high uncertainty, highlights the critical need for flexibility in power systems. Aiming to maximize the remaining flexibilities in power systems while ensuring network security, a novel concept of the Flexible Energy Market (FEM) is proposed. Leveraging closed-loop optimization of energy bids at the nodal level and network states derived from 5-minute state estimation data, the FEM optimizes generation, demand, and network constraints, facilitating additional secure energy exchange between fast units in real-time thus mitigating uncertainties in renewable energy production forecasts. Accurate determination of network security constraints, through a Common Grid Model (CGM), a list of Critical Network Elements and Contingencies (CNEC), and relevant reliability margins (RM), are crucial for enabling this exchange. A methodology for the determination of key network constraints for the close-to-real-time FEM, aiming to optimize the usage of flexible units in the electrical network, is developed. Furthermore, a software demonstration platform for real-time assessment of cross-zonal capabilities for energy exchanges was developed and demonstrations were performed with real network cases under the HORIZON 2020 OSMOSE project, which showed the feasibility of close-to-real-time energy exchanges even at congested borders.

1. Introduction

The rapid integration of renewable energy sources with a high level of uncertainty increases the need for flexibility in power systems (Harker Steele et al., Apr. 2021; Corona et al., Nov. 2022). Its critical role in enabling the successful integration of renewable energy into power systems while ensuring a reliable and efficient electricity supply is emphasized (Impram et al., Sep. 2020). Despite moving closer to real-time, current electricity markets close one hour before real-time, and flexibility reserves remain largely unused (Lam et al., Feb. 2018). In addition to this, considerable flexibility capacities prequalified and reserved for balancing mechanisms are not used for that purpose, waiting for a second chance (Krstevski et al., Nov. 2021). Transmission capacity calculation is inflexible in rapidly changing market and network situations (Gunkel et al., May 2020), even in the case of flow-based calculation, which retains zonal resolution (Henneaux et al., Jan. 2021; Kristiansen, Jan. 2020). Any change of zonal net positions affecting load flows may burden or relieve the network implying either market opportunities or inevitable remedial actions (Day-ahead capacity calculation methodology of the Core capacity calculation region, 2023). In this context, a flow-based method for allocating nodal costs in a European electricity network with high renewable energy penetration becomes very attractive (Vlachos and Biskas, Jul. 2021). By considering the spatial distribution of renewable generation and transmission constraints, costs associated with electricity flows at different network nodes would be accurately assigned as investigated in (Tranberg et al., May 2018). While various technologies and strategies have been explored to enhance flexibility (Babatunde et al., Feb. 2020), including the potential role of aggregators in managing grid congestion (Pantoš, Oct. 2020) and optimizing resource mixes for seasonal demand variations (Heide et al., Nov. 2010), there remains a significant gap in the development of a comprehensive and real-time flexible energy market solution (Pantoš, Oct. 2020; Leiva et al., Sep. 2023). To address this gap, this paper proposes a novel flexible energy market (FEM) concept, designed to optimize generation, demand, and network constraints while maximizing the remaining flexibilities within the power system. With an ambition of getting the most from the remaining flexibilities of the power system while maintaining secure network

* Corresponding author at: Electricity Coordinating Center (EKC), Belgrade, Serbia.
E-mail address: danka.veljkovic@ekc-ltd.com (D. Veljković).

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operation, the concept of a novel flexible energy market is proposed here, based on the closed-loop optimization of energy bids at the nodal level and network states determined from state estimation data at the 5-minute level. Such a market could also help deal with uncertainties in renewable energy production forecasts that are significantly lower coming closer to real-time.

For purposes of enabling additional secure energy exchange between fast units in real-time obtained from FEM, it is necessary to correctly determine network security constraints in a form of an accurate common grid model (CGM), a list of potentially limiting critical network elements and contingencies (CNEC), and relevant reliability margins (RM). Network assessment of parallel realities is performed within contingency analysis based on the determined 5-minute network snapshots from state estimation and a prepared list of constraints. FlexEnergy bids are provided based on internal portfolio optimization taking into account correlation with other commercial markets and commitment to the balancing mechanism. For accuracy purposes, bids are related to nodal positions, even if portfolio-based. The optimization engine performs closed-loop optimization of validated FlexEnergy bids and prepared transmission constraints while maximizing socio-economic welfare under the secure operation of a power system, (Cosson et al., Oct. 2020).

The key contribution of this work lies in the development of methods for defining critical network constraints essential for the operation of the close-to-real-time FEM, thereby facilitating enhanced utilization of flexible units connected to the electrical network. Additionally, this paper describes a software demonstration platform, the Electricity Network for Market (EN4M) tool, which enables real-time assessment of available cross-zonal capabilities for energy exchanges, developed under the OSMOSE project[1], (D6.2 Software demonstration platform development, 2023). Through simulations conducted in the region of Northern Italy and Slovenia, the EN4M tool provides valuable insights into the feasibility and efficacy of the proposed FEM solution (OSMOSE-D6.5-Demonstration-tests, 2023; OSMOSE-D6.6-Impact-analysis, 2023). In summary, this paper not only identifies the pressing need for a real-time flexible energy market solution but also presents novel methodologies and tools aimed at addressing this challenge, thus contributing to the advancement of power system flexibility and the integration of renewable energy sources.

2. The concept of the real-time preparation of transmission network data

A fundamental part of the proposed platform for the real-time preparation of network data is the automatic creation of merged transmission network models and an optimal list of critical network elements with contingencies. The network state is described through common grid models, which are created using the state estimation data with a 5-minute resolution. Each participating TSO should deliver snapshot models, produced by their state estimators, to the platform thereby providing raw data on active and reactive power flow, voltage, and angle of all nodes in the modeled high-voltage network.

To ensure secure network operation with optimized generation patterns obtained from the flex energy market, it is important to appropriately consider all potential contingency cases. All relevant contingency cases have to be defined in the form of pairs of critical network elements and contingencies, ensuring that all potentially critical network states will be assessed with contingency analysis. As the target electricity market is very close to real-time, the complete network model creation and energy market optimization process have to be fast enough to leave time for bidding and unit activation processes. Hence, all processes should be maximally optimized to cope with narrow deadlines imposed by the 5-minute time resolution of FEM. Time constraints are additionally strict in the case of cross-border trades. For this reason, the number of relevant network states to be assessed within optimization should be minimized by leaving only potentially limiting network states for the actual timestamp. The presolve algorithm is developed to avoid analysis of all possible contingencies by determining all redundant network constraints for that specific network conditions.

It is important to note that bids for fast-responding units that are aiming to participate in FEM are modeled with a nodal resolution to provide exact localization and accurate network representation and analysis. The complete FEM process is based on the nodal network representation, with no difference between internal and cross-zonal limitations.

The complete process of transmission network data preparation consists of the following steps:

1. Import and validation of the snapshot models produced by different state estimators.
2. Preparation of raw transmission network data of all nodes in the observability area.
3. Creation of a common grid snapshot model based on the newest state estimation data.
4. AC load flow calculation on the created model and validation of the results.
5. Adjustment of the created model to its further usage in the DC-based market calculation.
6. Final check of the created DC-ready model.
7. Flow-based capacity calculation of remaining available margin (RAM) and Power Transfer Distribution Factors (PTDF) matrix for every CNEC.
8. Optimization of the list of network constraints by applying the two-phase presolve algorithm.
9. Creation of a final list of non-redundant network constraints in an appropriate format for market calculation.

The described concept is designed and developed in the form of four modules integrated into the common platform for network data preparation, as illustrated in the high-level diagram shown in Fig. 1.

The first three modules include the functions for the simultaneous creation of CGMs based on updated state estimation data. The primary one takes the latest state estimation data from the server, validates it, and extracts information on generation, load, topology, shunts, and interchanges required for the model creation. The second one uses a previously validated common merged network snapshot model and real-time data files that are created periodically (every 5 minutes) based on the newest P, Q, V, Θ data from state estimators and thus creates the AC CGM for that timestamp. Finally, the created AC CGM is adapted to be ready for DC-based market calculation by converting calculated active power losses per branch into injections at the sending end and adjusting the maximum allowable current of branches by reducing the impact of corresponding calculated reactive power flows. After load flow calculation and validation from the point of calculation results and format, a final merged model is exported to the server in an appropriate format (CGMES, UCTE, or RAW).

The last module includes the functions for the preparation and filtering of the CNEC list. It integrates flow-based capacity calculation and two presolve algorithms that use all available data on network parameters and bids to optimize the CNEC list. A maximally reduced list of potentially limiting network constraints is delivered to the server for the FEM optimization.

For scalability reasons, all functions of the real-time network data preparation platform are designed to support different model formats,

[1] This work is part of the Optimal System-Mix Of flexibility Solutions for European electricity – OSMOSE project. It has received funding from the European Union's Horizon 2020 research and innovation program under grant agreement n°773406. This article reflects only the authors' views. The European Commission is not responsible for any use that may be made of the information it contains.

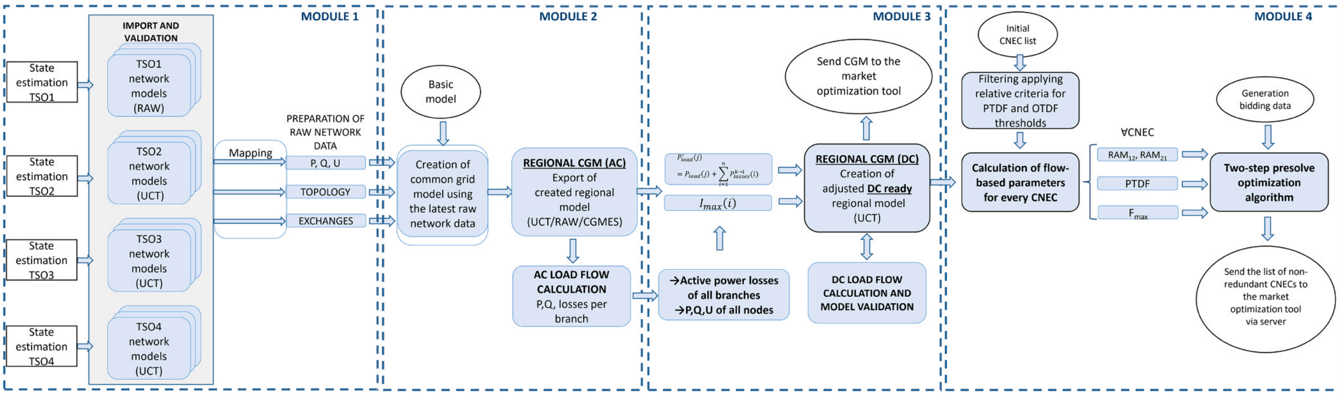

Fig. 1. Real-time network data preparation process flow.

time resolutions, and regions of interest.

3. Creation of a regional network model

Accurate network representation and analysis are of crucial importance for secure power system operation, which is why particular attention is paid to the creation of a realistic network model when ensuring safe network operation. The main basis for the creation of a network model for flex energy market optimization relies on real-time data obtained from power system state estimators. Individual snapshot models created by the SCADA system of each TSO using the latest available set of measurements are the main inputs for the creation of a common grid model. Discrepancies among formats of exported snapshot models by different TSOs imposed the need for a unified concept for the creation of CGM similar to congestion forecast procedures, where the basic snapshot model is fixed and variable data are taken in raw format and applied to that model. Variable data of interest are not forecasted data as in the case of congestion forecast procedures (such are two-day-ahead (D2CF), day-ahead (DACF), and intraday congestion forecast (IDCF)) but a set of measured data that are produced by the state estimation. Active and reactive power injections, voltage setpoints, topology statuses, transformers tap positions, and net positions are taken from different snapshot models delivered with a 5-minute time resolution.

While all factors are sources of uncertainty for the accuracy of flow-based market coupling, base case computation, and generation shift methods represent the two main deliberate aspects, as described in (Schönheit et al., Jan. 2020). A novel algorithm for computing injection shift keys, aiming to improve the accuracy and efficiency of power system state estimation processes, may offer benefits such as better estimation of power flow and system operating conditions, ultimately contributing to the reliable operation and management of power grids. (Van Den Bergh and Delarue, May 2016) The preference for a DC power flow representation over a non-linear AC power flow is highlighted in (Van Den Bergh and Delarue, May 2016) due to the significant computational cost associated with the latter, with the DC power flow providing a linear relation between power injections and power flows through power transfer distribution factors (PTDF), despite resulting in approximately 5 % average differences in power flows compared to AC models, particularly in high voltage grids, though deviations for individual lines may be larger. In (Kumar et al., Nov. 2004) an approach utilizing AC transmission congestion distribution factors to tackle zonal congestion within the network is introduced, potentially involving the computation and application of these factors to pinpoint congested zones and enact strategies for congestion relief, thus contributing to the advancement of efficient congestion management techniques in power systems. However, as explained in an evolutionary framework for transmission power systems with high renewable energy penetration (Zhang et al., Mar. 2017), a network simplification method based on PTDFs is introduced to reduce grid size complexity.

In contrast to all existing capacity calculation procedures, where the network is modeled applying the best available forecast on load, generation, topology, and exchange data, the innovative concept here applies closed-loop calculation with real-time network measurements gained from the power system state estimators. As illustrated in Fig. 2, the initially merged and validated snapshot model of the complete observability area of interest is updated every 5 minutes with the latest state estimation data delivered by corresponding TSOs. The regional model is adjusted by applying load distribution factors within an iterative process based on the load flow calculation results and the following Eq. (1):

$$P'_{load,net}(i) = P_{generation,net} - P_{exchange} - P_{losses}(i) \tag{1}$$

As energy market optimization is usually based on a linearized representation of the network, the final merged model is adapted for further usage in DC calculation. However, it is important to point out that formulations neglecting losses can have significant drawbacks and produce inaccurate results that overlook technical infeasibilities arising from overloadings, (Neumann et al., May 2022). Therefore, the initially created AC model is adjusted to be a DC-ready model and obtain power flows from the DC load flow closest possible to the ones from the AC load flow. Active power losses $P_{losses}^{k\to l}(i)$ are calculated in AC load flow calculation for each branch i connecting nodes k and l. Active power injections of each node from the initial model should be recalculated based on Eq. (2) taking into account calculated active power losses of each connecting element as injections at its sending end:

$$P'_{load}(j) = P_{load}(j) + \sum_{i=1}^{n} P_{losses}^{k\to l}(i), \tag{2}$$

$P_{load}(j)\lfloor MW \rfloor -$ active power injection at node j from the initial model,

$P'_{load}(j)\lfloor MW \rfloor -$ recalculated active power injection at node j,

$P_{losses}^{k\to l}(i)\lfloor MW \rfloor -$ active power losses of branch i connecting nodes k and l,

$n -$ number of connecting elements to the node j.

It is important to note that the estimated flows per border lines that have both sides modeled are not taken into account during the CGM creation, due to discrepancies between those flows from different sides, caused by different state estimators. Therefore, the related injections on paired X nodes are removed before the merging procedure and these flows are later calculated by performing load flow calculation on the merged model. Fig. 3

The second adjustment of the created model is related to the modification of the maximum permanent allowed current $I_{\max}(i)$ of branches to quantify the influence of the calculated reactive power flow $Q_{calculated}^{k\to l}(i)$. $I_{\max}(i)$ is reduced for the influence of reactive power flows $Q_{calculated}^{k\to l}(i)$ by taking into account the appropriate power factor, as

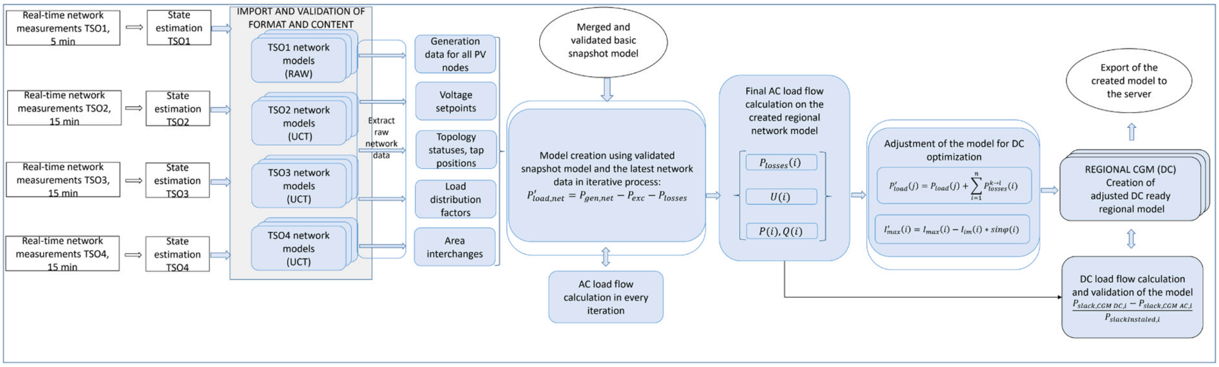

Fig. 2. Creation of Regional CGM.

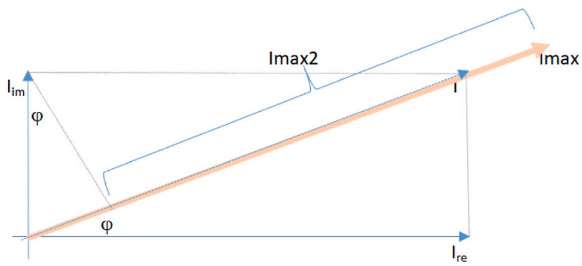

Fig. 3. Illustration of maximum line current adjustment for DC calculation.

described in Eqs. (3) and (4).

$$I'_{max}(i) = I_{max}(i) - \frac{Q_{calc}^{k \to l^2}(i)}{\sqrt{3} \quad U(i)\sqrt{\left(P_{calc}^{k \to l^2}(i) + Q_{calc}^{k \to l^2}(i)\right)}} \tag{3}$$

$$I'_{max}(i) = I_{max}(i) - \frac{Q_{calc}^{k \to l}(i)}{\sqrt{3} \quad U(i)} * \sqrt{1 - \cos^2\varphi(i)} \tag{4}$$

Finally, a validity check is performed by comparing the power flows obtained from the initial AC load flow calculation, with those from the final DC load flow calculation using the modified model (D6.4, 2023). To ensure that market optimization is run only on consistent networks, the amount of power imbalance (MW) in the DC snapshot per timestamp is calculated as follows:

$$KPI_{DC,mismatch} = \frac{P_{slack,CGM\ DC,i} - P_{slack,CGM\ AC,i}}{P_{slackInstaled,i}} \tag{5}$$

$P_{slack,CGM\ DC,i}$ – injected active power at slack node i obtained from DC LF calculation,

$P_{slack,CGM\ AC,i}$ – injected active power at slack node i obtained from AC LF calculation,

$P_{slackInstaled,i}$ – installed power at slack node i.

4. Determination of the optimal list of constraints

Determining the right set of network constraints is one of the key factors for network security. Network constraints are defined in the form of a list of critical network elements and contingencies (CNEC) and taken into account by the market coupling algorithm (Schönheit et al., May 2021). An optimal list of constraints should be determined to keep the balance between ensuring secure system operation in case of contingency and reducing optimization time as much as possible.

Assessment of the optimal list of constraints valid for the specific timestamp is a major concern for such a time-demanding energy market, due to its narrow deadlines, and should be consisted of the following steps:

1. National CNEC lists should be formulated by each TSO, who is the main responsible for maintaining the network security, and afterward merged into one common CNEC list
2. The initial common CNEC list is afterward filtered by applying relative thresholds of zonal PTDF (5 % for critical network elements, according to the Core FB methodology) and OTDF (e.g. 10 % for contingencies)
3. The filtered CNEC list is additionally optimized through the two-step presolve algorithm leaving only non-redundant CNECs, potentially limiting for a given specific network and market state.

4.1. Presolve algorithm

Presolve is a constraint optimization step that ensures the determination of a moderate number of CNECs for the market optimization algorithm and thus increases its performance and reduces the execution time. Presolve analysis is developed based on two different algorithms, where the first one uses a zonal approach, while the second one applies a nodal approach, but can be also used as zonal.

Presolve analysis follows the initial centralized flow-based capacity calculation step and takes the following calculated classes of parameters for each CNEC:

- Remaining available margin in both directions (RAM_{12}, RAM_{21})
- Nodal and zonal power transfer distribution factors ($PTDF_{nth}$, $PTDF_{zth}$)
- The maximum allowed capacity of the network element (F_{max})

The Remaining Available Margin is calculated for both directions of each CNEC (6) by reducing the maximum capacity of critical network element with its initial non-commercial flow and flow reliability margin, following Core flow-based definitions:

$$RAM = F_{max} - F_0 - FRM \tag{6}$$

$$F_{maxDC} = \sqrt{3} \bullet U \bullet I_{maxDC} \tag{7}$$

$$F_0 = F_{ref} - [PTDF] \times [BCE] \tag{8}$$

$F_{max}[MW]$ – Maximum capacity of the element
$F_{ref}[MW]$ – Active power flow in the base case
$F_0[MW]$ – Loop flow in the base case
$FRM[MW]$ – Flow Reliability Margin
$BCE[MW]$ – Base case exchanges among participating areas
$I_{maxDC}[kA]$ – Maximum DC-ready allowed current, calculated as described in Chapter 3.

Nodal (node-to-hub) power transfer distribution factors ($PTDF_{nth}$) illustrate the sensitivity of a particular line to the injection change at a relevant node. Nodal PTDFs are determined by changing the injection of a given node from the CGM and calculating the relative power flow difference on each CNEC.

Zone-to-hub PTDFs are calculated based on the nodal PTDFs (9) by multiplying the proportion of each node's contribution in the GSK with the corresponding nodal PTDF and summing up these products, according to the following equation:

$$[PTDF_{z2h}] = [PTDF_{n2h}] * [GSK_{n2z}] \tag{9}$$

$PTDF_{z2h}$ – matrix of zone-to-hub PTDFs
$PTDF_{n2h}$ – matrix of node-to-hub PTDFs
GSK_{n2z} – matrix containing the GSKs of all bidding zones

The zone-to-hub power transfer distribution factors ($PTDF_{zth}$) indicate how a change in the bidding zone net position affects the power flow on each CNEC, calculated based on the previously created CGM and the defined GSK. To evaluate the influence of exchanges between different zones $A \rightarrow B$ on the respective CNE, zone-to-zone PTDFs ($PTDF_{A\rightarrow B,CNE}$) can be obtained from the zone-to-hub PTDFs as follows:

$$PTDF_{A\rightarrow B,CNE} = PTDF_{Ath,CNE} - PTDF_{Bth,CNE} \tag{10}$$

The first part of the optimization algorithm is based on a comparison of the power flow sensitivity of all CNECs to interchanges between all participating TSOs. The basic idea of this algorithm is to check whether the RAM reserves of some CNECs will always be higher than the other CNECs for all combinations of interchanges. The second part of the optimization process relies on exploring whether power flows resulting from all possible combinations of real generation limits of nodes/zones may fill out the capacity of CNEC. All CNECs that are not critical in any combination of interchanges are marked as redundant and should not be taken into account as a constraint in the market optimization process.

4.2. Initial sensitivity presolve algorithm

The initial part of the presolve algorithm calculates the sensitivity of power flow changes induced by interchanges between participating bidding zones. Change in power flow is calculated for each CNEC for all combinations of interchanges with appropriate zonal PTDF values calculated for that specific CNEC. These values are kept in an array and used as a weighting factor, as illustrated in the following figure. Fig. 4

RAM values for forward and reverse direction and weighting factors are first calculated for every CNEC and then checked against redundancy conditions. CNEC is determined as redundant if there is another CNEC that fulfills the following criteria:

1. Weighting factors (WF) have the same sign for all combinations of interchanges (A→B, A→C, …, C→D)
2. $WF_{X\rightarrow Y,\,1} \geq WF_{X\rightarrow Y,\,2}$ in case of both non-negative WFs
3. $WF_{X\rightarrow Y,\,1} \leq WF_{X\rightarrow Y,\,2}$ in case of both negative WFs
4. $RAM+_{CNEC2} \geq RAM+_{CNEC1}$
5. $RAM-_{CNEC2} \geq RAM-_{CNEC1}$

The listed conditions imply that power flow for $CNEC_1$ will always reach its RAM limits faster than $CNEC_2$, meaning that for all potential combinations of exchanges, the impact on $CNEC_1$ would be greater (or at least equal) than the impact on $CNEC_2$. Consequently, remaining available margins will be first breached for the first CNEC and thus it should be treated as non-redundant. The calculation principle is illustrated in the simple example with four bidding zones and presolve calculation parameters are compared for 4 CNECs in Table 1.

From the table it can be seen that limits for $CNEC_1$ will always be reached faster when compared to $CNEC_2$, meaning that RAM reserves in both directions for $CNEC_2$ are higher than for $CNEC_1$ for the same combination of interchanges. To illustrate this example, it can be said that $CNEC_2$ is always covered by $CNEC_1$, while this is not the case with $CNEC_3$ for which the weighting factor for interchange A→B is higher than the one for $CNEC_1$, resulting in a discrepancy with the second condition. In the case of $CNEC_4$, it can be seen it is not covered by $CNEC_1$ due to opposite signs of corresponding weighting factors for interchange A→B, which is the first redundancy condition.

The applied presolve algorithm enables tracking of the redundancy relations between CNECs and results may be used in the preparation phase of a list of constraints for the selection of the presolved CNECs, which can potentially be limiting, at the same time simplifying the optimization problem by removing obsolete, redundant constraints. If there is a CNEC, which due to the grid configuration for all combinations of interchanges always "covers" another CNEC, then the second one should not be considered when creating the initial CNEC list.

4.3. Final presolve algorithm for the actual state

The second part of the presolve algorithm takes either the range of net positions of bidding zones in the case of a zonal application or real generation limits of nodes for a nodal application as an additional restriction. It aims at determining all potential network constraints that could be critical in a realistic domain for that specific timestamp. For these purposes, TSO should provide realistic ranges of net positions for each bidding zone or generation limits for each production node that may be varied in the timestamp of interest.

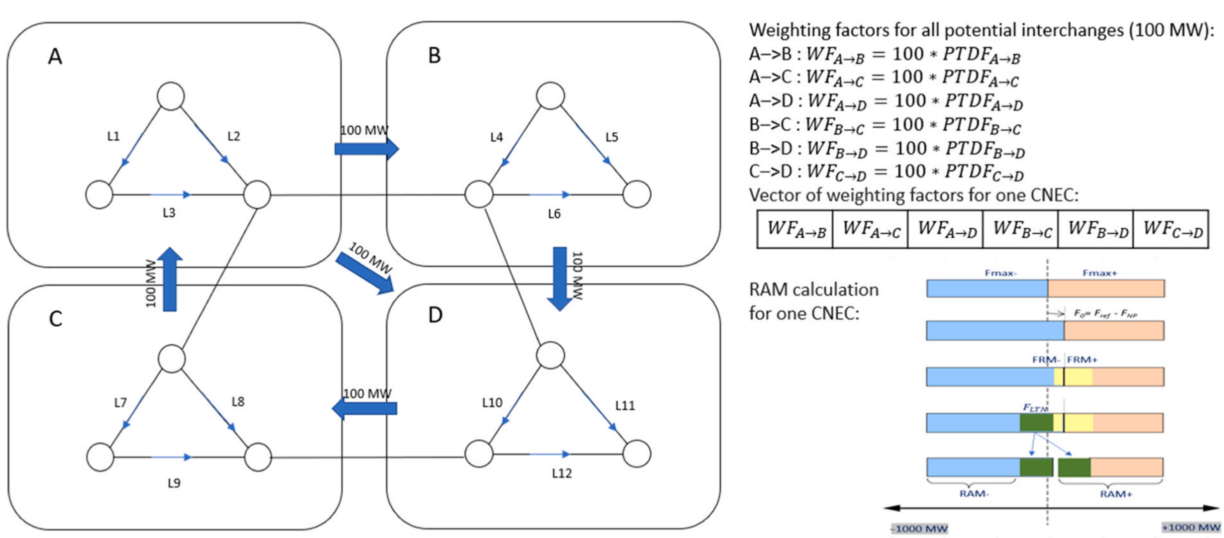

Fig. 4. Illustrative example of calculation of initial presolve parameters in case of 4 bidding zones.

Table 1
Example of calculation of weighting factors for 4 CNECs.

CNEC	CNE	Contingency	RAM_{12}	RAM_{21}	Impact of area interchange on CNEC					
					A→B	A→C	A→D	B→C	B→D	C→D
1	Line 1	Outage 1	1000	500	23	13	21	20	13	-17
2	Line 1	Outage 2	1100	600	18	8	12	9	7	-12
3	Line 2	Base Case	1100	700	27	6	14	11	6	-15
4	Line 3	Outage 2	1100	600	-2	8	18	7	12	-12

The algorithm is designed to exhaust the domain of feasible solutions and check whether the worst possible combination of interchanges in zonal or power injections in a nodal case may fill out the remaining available margin of any CNEC. Cumulative power flow is calculated for every combination of zonal/nodal positions in the given range for both directions of each CNEC and compared with forward and reverse RAM values:

$$\sum_{i,j=1}^{n} PTDF_{i-j} * (NP_{\max,i} - NP_{\min,j}) < RAM, \text{ in the case of zonal definitions} \tag{11}$$

$$\sum_{i,j=1}^{n} PTDF_{i-j} * (P_{gen,\max,i} - P_{gen,\min,j}) < RAM, \text{in the case of nodal definitions} \tag{12}$$

If the calculated cumulative power flow for specific CNEC is always lower than RAM_{12} and RAM_{21} that CNEC is considered redundant, as it will never breach its limits in the range of feasible solutions.

Presolve 2 is based on the greedy algorithm with two nested loops, where the first one looks for the most critical combination of zonal/nodal positions that maximally increase the power flow on the critical network element, while the second one checks the complete realistic range of positions provided by TSO.

Reduction capabilities of the second presolve algorithm are highly dependent on the size of provided net position/generation ranges in the way that a narrower range implies higher reduction. This is especially emphasized in closer to real-time energy markets, where the generation limits are close to the operating point and a realistic range can be precisely defined. This can be particularly important for improving the performance of the market optimization algorithm by reducing the number of non-redundant constraints and thus optimization time. Fig. 5

In OSMOSE project, the presolve algorithm is used within the online pre-processing tool for reducing the number of CNECs used in the FEM optimization process. Based on the real energy bids collected from the providers for that specific timestamp, realistic generation ranges of the corresponding nodes in the model are precisely determined and used by the second presolve algorithm. This enables the optimization of the flex energy market in the shortest possible time thus facilitating the application of the flex energy market closer to real-time.

5. Assessment of the reliability margin

Even though the flex energy market is using the latest available network data from state estimators and its time resolution is very close to real-time which enables a high quality of the forecasted network model, it is still necessary to define a certain reliability margin for ensuring secure operation of the power system. Moving closer to real-time significantly reduces the size of the reliability margin that is sufficient to cover all uncertainties that may occur between the time of capacity calculation and the real-time energy exchange. However, it also decreases the time to perform remedial actions and cope with unexpected grid situations. The reliability margin should be defined per critical network element as the amount of physical flow that should be reduced from the line capacity. The definition of this margin is in line with the Core flow-based calculation concept for the Flow Reliability Margin (FRM).

Following the recommended approach in this methodology, the flow reliability margin should cover all unintended deviations due to load frequency control and all uncertainties that could occur between the capacity calculation time unit and real-time and thus impact the reliability of capacity calculation. In the case of the flex energy market, which is the separated process from the real-time load-frequency control, the component of the margin that is related to unintended deviations of power flows remains as it is. The second component covering uncertainties is related to errors caused by network modeling and data forecasting. As this is the first energy market based on the closed-loop network data from state estimators with a very short period from calculation to operation time, the size of the margin should be minimal. The uncertainty component of the reliability margin is impacted by the inaccuracy of network data from state estimators and discrepancies between different state estimation techniques and formats of exported

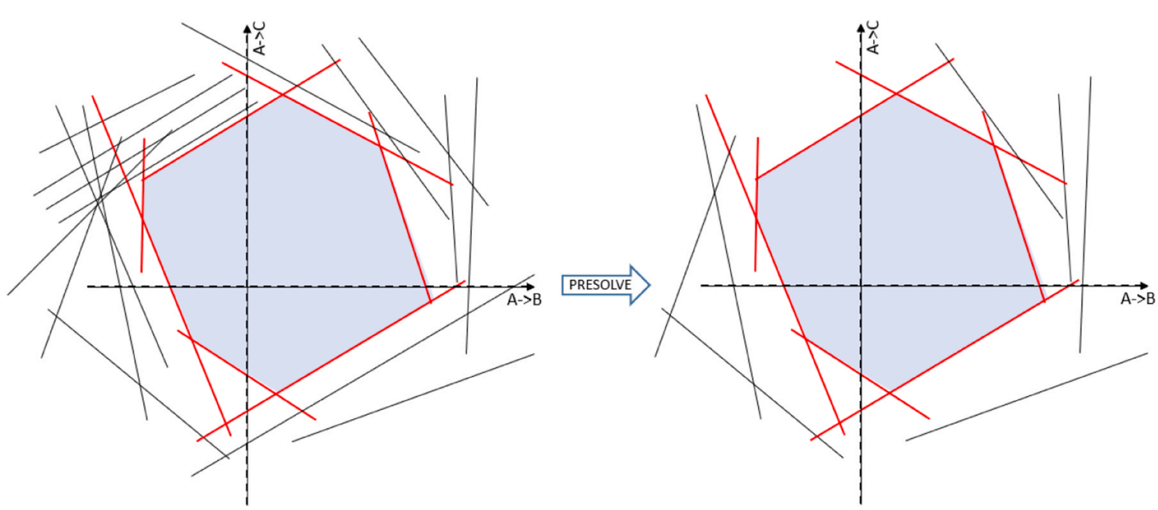

Fig. 5. Flow-based domain and presolve reduction capabilities for example of three areas.

data. This aspect (as well as the preparation time for the CGM) could be improved by having a common state estimation model at the regional or European level. Even though the nodal representation of a high-voltage network minimizes the forecasting error, a factor that may cause inaccuracies is the precision of the presentation of generation share among units in case of portfolio bidding.

The exact size of the flow reliability margin per CNEC should be determined based on the statistical assessment of differences between calculated forecasted and corresponding recorded realized flows, (Vujasinović et al., Jun 1, 2019). Forecasted flows should be calculated based on the adjusted network models with the realized data that do not impact the reliability margin. For this assessment, it is of major importance to create a relevant database for the statistical analysis and cover different seasons and sufficient periods. In the case of the OSMOSE project, demonstration tests were performed during short periods, which are not relevant for the exact determination of the value of the reliability margin. Additionally, no outages occurred during demonstration periods, and respecting contingencies couldn't be analyzed resulting in flow difference calculation per CNE instead of relevant CNECs. FRM value will be accessed from differences between flows for each analyzed timestamp that are calculated as follows:

$$\Delta F_i = \frac{F_{ref} - F_{realized}}{F_{\max}} \tag{13}$$

The size of the reliability margin varies around the value of a few % of the installed capacity of the line, which is significantly lower than the size of the reliability margin in the case of the day ahead market which usually takes at least 10 % of a CNE's installed capacity.

6. Demonstration of close to real-time cross-border flexibility at the Italian-Slovenian border

The complete process of the proposed flex energy market was simulated and demonstrated on the use case of observability area of Slovenian and northern Italian networks and 15-minute time resolution. A common network model is created using state estimation data from Slovenian (ELES), Italian (TERNA), and French (RTE) Transmission System Operators. It models the complete observability area of ELES and TERNA including neighbouring systems, as illustrated in Fig. 6. It is important to note that software is demonstrated and tested using real-time data for this use case, but is not limited to any network or time resolution described by this use case, meaning that it can be used for the simulation of any network from the size of IGM to entire Continental Europe or any other synchronous area as well as any time resolution from 1 to 60 (or more) minutes. Hence, the complete process is fully configurable and the area of interest and time horizon may be easily adjusted by defining a proper set of input data and parameters, Fig. 7,8

During the two-year period (June 2020 – March 2022), different types of tests were performed, from open-loop tests of individual parts of the platform to complex fully integrated real-time closed-loop tests. Open-loop tests included network, bids, and activation signals and checked everything in the sequence: from the snapshots provided by TSOs and processed by the network tool, to the activation list imported

Fig. 6. Observability area.

to SCADA and sent to flexibility providers. Final closed-loop tests aimed at testing and tuning integrated platform parameters under a real system environment considering coordinated real-time activation of three participating generation units. Proof of concept was successfully performed on 3.3.2022. resulting in the actual activation of generation units after submission of flexibility bids (arising as the result of generation portfolio optimization) to the FlexEnergy Market Platform aligned and approved by the optimization tool taking into account current network states and constraints determined by EN4M.

Different TSOs' snapshot creation options and procedures required the use of some backup procedures in comparison to the proposed methodology, such as the use of a 15-minute time resolution instead of the originally planned 5-minute resolution. Besides timing constraints, the quality of the merged network model is directly influenced by the individual snapshot models provided by each TSO. The overall time required for network data processing and preparation, followed by delivery of the DC-ready CGM and related presolved CRAC list, typically takes 110–120 seconds.

The original CNEC list delivered by TSO was initially reduced by applying a relative 5 % PTDF and 10 % OTDF thresholds and contains 435 CNECs, including 93 critical network elements and 56 different topologies. When processing the actual network data for every timestamp, this initial list of constraints is additionally reduced by applying two presolve algorithms in the EN4M tool.

The analysis of network data generated during demonstration tests suggests that a considerable number of constraints may be omitted, hence accelerating the market optimization process. Fig. 9 depicts the percentage of the initially presolved CNECs, while Fig. 10 shows the total number of redundant CNECs determined by the first and second optimization algorithms throughout analyzed timestamps during the demonstration period. While results are primarily presented from official demonstrations, it is important to note that during the two-year testing and demonstration period, the FEM is rigorously tested across various network states to assess its performance under different conditions and congestion levels. Simulations covered peak and off-peak network states across all day and night hours in both summer and winter seasons, ensuring a comprehensive assessment of the FEM's performance. For clarity reasons, results are presented only for typical regimes showing the total number of presolved CNECs for winter network regimes based on January simulations in Fig. 11 and for summer network regimes based on July simulations in Fig. 12. The results demonstrate that in more congested scenarios, the presolve algorithm effectively reduces a greater number of constraints.

The number of presolved CNECs that can potentially be limiting and should be considered in the market optimization process is shown in Fig. 11. It is necessary to emphasize that the activations of three flexibility units during demonstration tests weren't large (two 5 MW/5 min and one 10 MW/5 min) and such a narrow generation range applied within the second presolve algorithm didn't leave a significant number of potential security constraints.

7. Discussion on the practical implementation

The demonstration tests proved that even at the most congested borders, such as the border between Italy and Slovenia, close to real-time energy exchanges can be achieved promptly based on the latest grid snapshots and close to real-time bids. It needs to be highlighted that flexibility trade is not limited to cross-border or cross-zonal interchange but can also occur inside a certain bidding zone because the nodal model of the network is applied.

In terms of existing energy market infrastructures, the FEM is expected to be compatible with the current infrastructure of the Automatic Frequency Restoration Reserve (aFRR) and should be able to coexist, using the aFRR channels to communicate and run the dispatchable flexibility sources.

From the network modeling perspective and its real-time

Fig. 7. ENTSO-E map of part of the demonstration network.

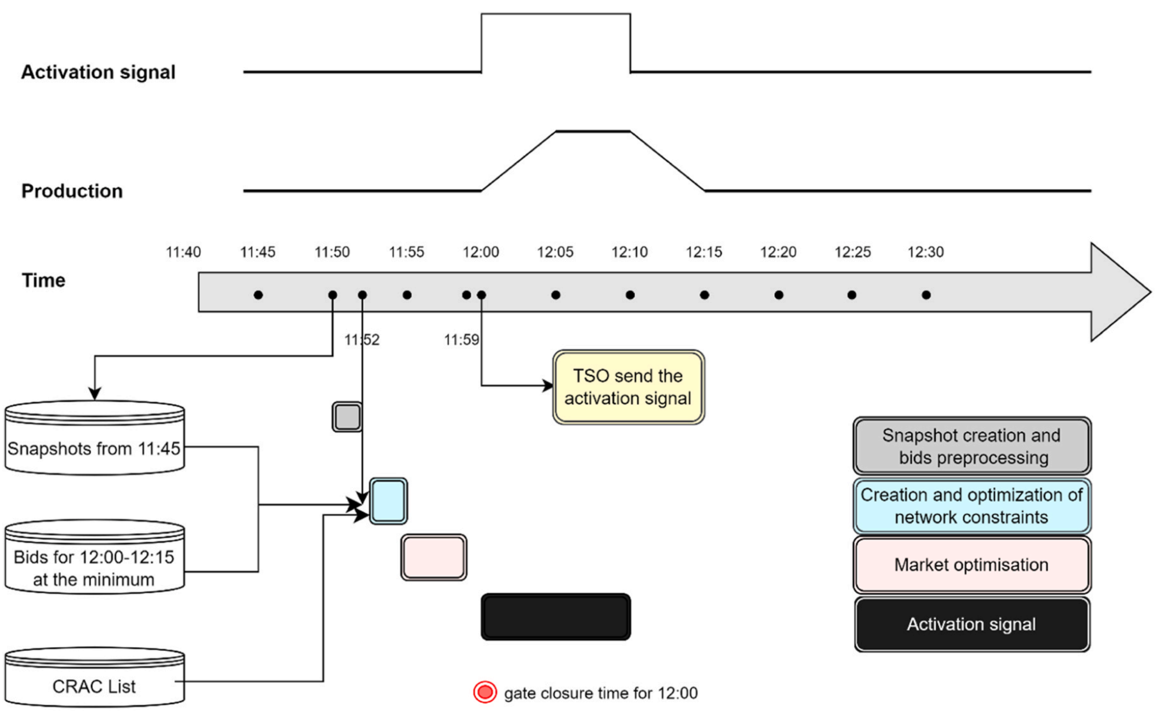

Fig. 8. High-level diagram of flex-energy market demonstration.

representation, the lessons learned so far show that for close to real-time markets such as the FlexEnergy one, the quality of state estimation and the speed of providing and merging network data snapshots are essential for the quality and performance of the process. Regarding the network constraints, the FEM relies on the ability to swiftly produce network snapshot models from TSOs' state estimators. This function exists in every TSO but as observability areas typically cover their own TSO and some surrounding network, it requires merging of the individual snapshot to create a common (e.g. continental European) snapshot model. This merging takes some time, delays the process, and brings inaccuracy in modeling. Hence, a common synchronous area-wise state estimation is seen as needed and is recommended as one of the future developments. Such a function would also provide benefits to the balancing timeframe capacity calculation, provided in the EC's Electricity Balancing Guidelines, Article 37(3), Commission Regulation EU, 2017. In such a case the time-consuming process of merging individual models would be avoided; full synchronicity and data consistency would be achieved. Such a procedure (for instance based on updating variable CGMES data) would ensure pan-European snapshots in a few seconds, ready for the needs of optimization at close-to-real-time.

To ensure computational efficiency and scalability of the closed-loop optimization algorithm for the FlexEnergy Market (FEM), several

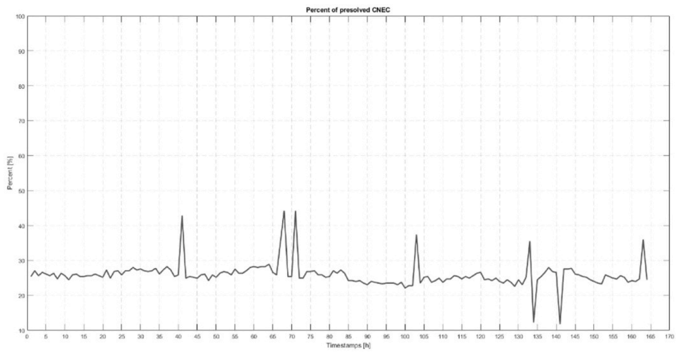

Fig. 9. Percent of the initially presolved CNECs.

measures have been implemented and tested over a two-year project period. These include process parallelization, fallback procedures, and a modular architecture. Timing discrepancies necessitated software adjustments, with a combination of event-triggered and time-triggered procedures implemented to ensure continuous operation. However, scaling up with additional flexibility units may require improving computational efficiency through software and hardware enhancements. Potential challenges may arise with scalability and timing performance, particularly with a higher number of providers and bids, which could be addressed in a follow-up project. Technical barriers to scaling mainly involve upgrading infrastructure to meet requirements, particularly regarding interoperability and activation processes.

The socio-economic analysis leads to the estimation that the economic potential for such a market is now rather low in comparison to other trends in the power market. Meanwhile, as more uncertainty arises in real-time as a result of the penetration of renewable energy sources into electrical power systems, such flex-energy markets may become more valuable commercially in the years to come. This market can be used for compensating RES production forecasting error as the RES uncertainty is significantly lower coming close to real-time.

8. Conclusions

This paper investigates the potential of near real-time flexibility of power systems and energy markets in terms of the remaining transmission capacity that can be exploited. It introduces and describes methodology and procedures for enabling the proposed flex-energy market, emphasizing the speed and robustness from a grid security perspective, as transmission operators' capabilities to act beyond this phase and take corrective measures are limited. Three calculation engines were developed and incorporated into software demonstration platforms to facilitate real-time network assessment, energy bid creation, and market optimization.

The research focuses on accurately identifying transmission network

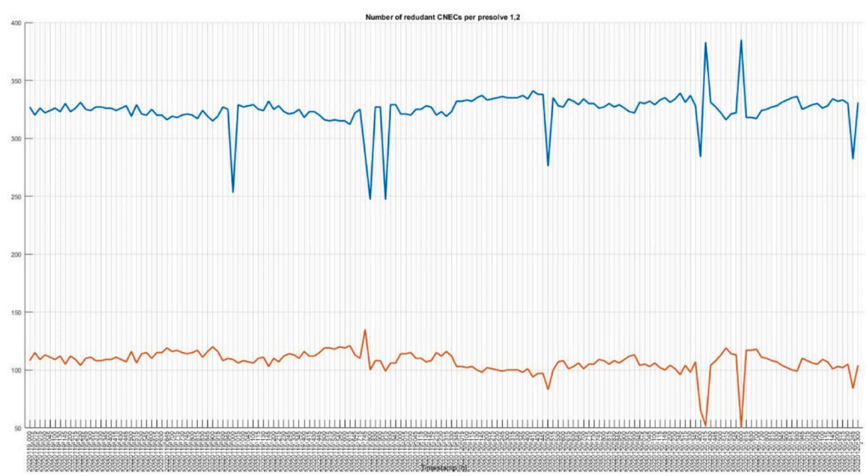

Fig. 10. Number of redundant CNECs determined by each presolve algorithm.

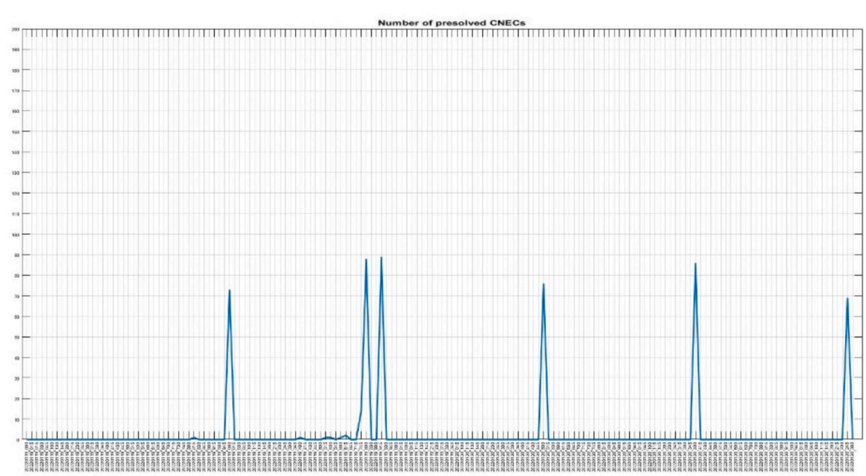

Fig. 11. Total number of presolved CNECs for winter network regimes.

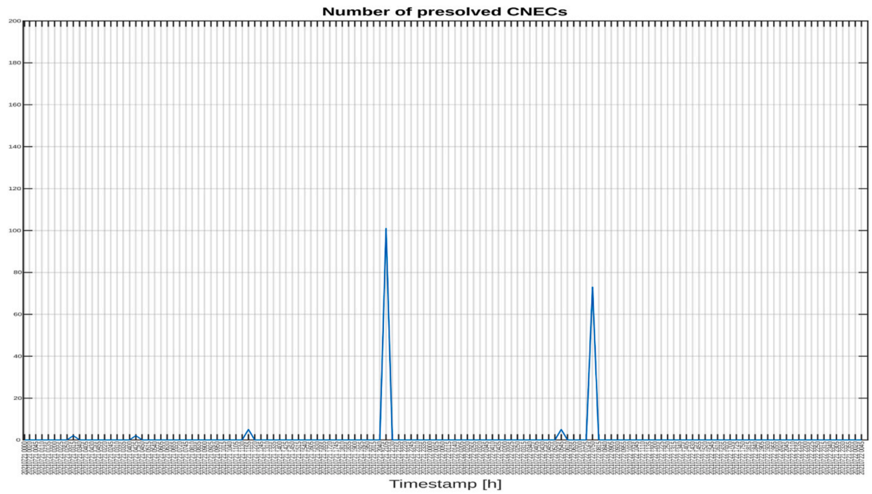

Fig. 12. Total number of presolved CNECs for summer network regimes.

constraints and evaluating cross-zonal energy exchange capabilities within a 5-minute timeframe using state estimation data. It proposes and describes algorithms and techniques for assessing the most impacting transmission network constraints in the form of an accurate network model and a list of non-redundant CNECs based on the current network state represented by the most recent real-time measurements. Results of the calculations show that 70–80 % of the predefined network constraints are reduced by proposed algorithms speeding up the calculation process significantly.

Over a two-year period, the proposed procedures were tested, addressing challenges in data consistency and format across transmission system operators (TSOs). Efficient and robust network representation requires the implementation of a pan-European single-state estimation process and synchronized network snapshot modeling, thus minimizing the need for merging individual models. Demonstration tests show the feasibility of close-to-real-time energy exchanges, even at congested borders.

From a socio-economic perspective, even though the current economic potential of such markets is low compared to other power market trends, uncertainties arising from renewable energy source penetration could increase their commercial value in the future so they could also serve to compensate the renewable energy production forecasting errors.

CRediT authorship contribution statement

Zeljko Djurisic: Writing – review & editing, Supervision. **Milan Ivanovic:** Validation, Software. **Zoran Vujasinovic:** Writing – review & editing, Project administration, Conceptualization. **Danka Veljković:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that has been used is confidential.

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